Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/05/2012
PI Name:	Ploutz-Snyder, Lori L. Ph.D.		
Project Title:	Integrated Resistance and Aerobic Training Study (Sprint)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ann Arbor	**State:**	MI
Zip Code:	48109-2013	**Congressional District:**	12
Comments:	Previously at Universities Space Research Association/NASA Johnson Space Center until July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2009	**End Date:**	09/30/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Project at JSC ended and continued by Integrated Resistance and Aerobic Training Study (Sprint)--NNX16AO74G, per HRP and PI (Ed., 10/30/18) NOTE: change in end date to 5/14/2021 per HRP MTL 11/11/11 (Ed., 12/22/11) NOTE: project title changed to "Integrated Resistance and Aerobic Training Study (Sprint)" per PI/JSC (7/2010) NOTE: change in end date to 6/1/2020 (from 9/30/15) per JSC (3/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Trappe, Scott (Ball State University) Ploutz-Snyder, Robert (USRA) Guilliams, Mark (Wyle) Martin, David (Wyle)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Current exercise countermeasures are insufficient to prevent muscle atrophy, cardiovascular deconditioning and bone loss associated with long-duration spaceflight. A known limitation has been the inability of the ISS exercise hardware to provide sufficient loads to the human body that are required for maintaining physiological function. New flight exercise hardware, including the advanced resistance exercise device (ARED) and 2nd generation treadmill (T2) are designed to provide astronauts with the ability to exercise at higher intensity. This opens an array of new possibilities for exercise programming in long-duration spaceflight. While the ability to do resistance exercises at heavier loads, the ability to run faster and potentially with more body loading are obvious improvements in exercise capabilities, the details of how this new equipment should be used are not so obvious. Towards this end, two workshops were held, the NASA Muscle Workshop in June 2008 and the NASA International Space Station Exercise Prescription Workshop in October 2008. Intramural and extramural experts concluded that using higher intensity resistance exercises and interval aerobic exercise would help to maintain physiological function while simultaneously decreasing total exercise time and volume. The panel then recommended that this evidence-based approach to the development of a novel exercise prescription for use on ISS should immediately be proposed. Therefore the purpose of this research is to evaluate the efficacy of a new integrated resistance and aerobic training (iRAT) program designed to minimize loss of muscle, bone, and cardiovascular function during ISS missions. Highlights of the iRAT include an increase in the intensity and a reduction in the volume of resistance exercises, inclusion of very short, but high intensity interval-type aerobic exercises, and starting the exercise countermeasures as early as possible in the flight, preferably in the first week. Pre-, in- and post-flight testing, and data sharing with selected on-going medical assessment tests, will be used to assess the effectiveness of this candidate prescription. Tests include detailed measurements of lower body muscle structure and function (MRI, ultrasound, muscle performance testing, shared muscle medical tests), maximal aerobic capacity, ventilatory threshold, left ventricular mass and cardiac contractility and bone mineral density. Twenty (20) long duration crew members will be recruited as active subjects in iRAT; and 20 will be asked to serve as controls that perform the standard care exercise prescription and the pre-post flight testing. This proposal is sponsored by the Exercise Physiology and Countermeasures Project and directly addresses the Human Research Program (HRP) Integrated Research Plan (IRP) Risks of Impaired Performance Due To Reduced Muscle Mass, Strength and Endurance and Risk of Reduced Physical Performance Capabilities Due To Reduced Aerobic Capacity and Risk of Accelerated Osteoporosis. This proposal addresses Gap M7: Can the current in-flight performance be maintained with reduced exercise volume? The proposal addresses IRP Gap M8 (What is the minimum exercise regimen needed to maintain fitness levels for operationally relevant tasks?) and Gap M9 (What is the minimum set of equipment needed to maintain those (M8) fitness levels?).
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Development of more efficient exercise programs could be used on Earth.
Task Progress:	Pre-flight baseline data collection was collected for the first ISS crew member participant who launched in the summer of 2011. During this reporting period the first in-flight exercise sessions were implemented, new communication tools developed for monitoring in-flight exercise, and the first in-flight muscle ultrasound scans were obtained.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Ploutz-Snyder L. "Gary A Dudley Memorial Lecture: Plasticity of Skeletal Muscle at the Extremes: Spaceflight to Elite Performance." 2010 NSCA (National Strength and Conditioning Association) National Conference, Orlando, FL, July 14-17, 2010. 2010 NSCA (National Strength and Conditioning Association) National Conference, Orlando, FL, July 14-17, 2010. , Jan-2010
Abstracts for Journals and Proceedings	Ploutz-Snyder L. "Exercise countermeasures for the maintenance of fitness during long duration spaceflight." American College of Sports Medicine 57th Annual Meeting, Baltimore, MD, June 2-5, 2010. American College of Sports Medicine 57th Annual Meeting, Baltimore, MD, June 2-5, 2010. , Jun-2010
Abstracts for Journals and Proceedings	Ploutz-Snyder L. "In-Flight Exercise – SPRINT." 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Ploutz-Snyder L. "Exercise Counteremeasures on the ISS: Integrated Resistance and Aerobic Training." Current Topics In Space Medicine Lecture Series, Cleveland Clinic, Cleveland, OH, August 2011. Current Topics In Space Medicine Lecture Series, Cleveland Clinic, Cleveland, OH, August 2011. , Aug-2011
Abstracts for Journals and Proceedings	Ploutz-Snyder L. "Candidate Exercise Technologies and Prescriptions." Workshop on Excercise Countermeasures. Life in Space for Life on Earth, Trieste, Italy, June 13-18, 2010. Workshop on Excercise Countermeasures. Life in Space for Life on Earth, Trieste, Italy, June 13-18, 2010. , Jun-2010